

Where Does Your Data Live?

Lecture Notes — Week 1: Pointers, Dynamic Memory, and Simple Contracts

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August 11, 2026

1.1 Why Data Organization Is the Subject

Read alongside: Horowitz & Sahni, *Fundamentals of Data Structures*, Ch. 1 (chapter-level); Aho, Hopcroft & Ullman, *Data Structures and Algorithms*, Ch. 1 (chapter-level).

You already know how to write a loop, a function, and a `struct`. It is worth asking, then, why one program answers a query instantly while another, computing exactly the same answer, takes an hour. The difference is almost never in the loop. It lies in how the data was organized before the loop ever ran. This course is a catalogue of those arrangements and, more importantly, of the judgement to choose between them.

Consider a concrete instance. A college stores student records, and given a roll number we must return the corresponding student. Suppose the records sit in an array, in the order the students were admitted. The obvious solution is to scan from the beginning until the roll number matches: about ten lines of C, correct on the first attempt. With a thousand students the query is instantaneous. With ten million records and a new query arriving every millisecond, that same correct program is unusable. Nothing about the loop was wrong; the arrangement of the data was wrong. By Week 8 we will answer the same query in roughly twenty-four comparisons rather than ten million, and by Week 10 in roughly one.

This is the gap the course exists to close, and it is worth being precise about why your existing training does not close it.

Definition (Structured programming). *The discipline of building programs from sequence, selection, and iteration, decomposed top-down into functions, avoiding unrestricted jumps.*

Structured programming gives us programs we can read: each function does one thing, and control flow can be traced by eye. But it disciplines only the *control* half of a program. It says nothing whatever about how the data those functions share is arranged. A program is algorithms operating on data structures; structured programming disciplines the first term, and this course disciplines the second [1].

Definition (Data structure). *A way of arranging data in memory, together with the operations that arrangement makes cheap.*

Two questions follow, and every structure studied in Weeks 3 through 12 will be judged on exactly these two axes. The first asks where the data lives: who allocates the memory a structure occupies, how large is it, and who is responsible for giving it back. That question is the subject of the rest of this chapter. The second asks what an operation on the structure costs — not “it felt fast,” which is not an argument, but a number expressed as a function of the amount of data. That question is the subject of Week 2.

The course answers these two questions over twelve weeks. Weeks 1 and 2 ask where data lives and what an operation costs — pointers, memory, and complexity. Weeks 3 through 5 take up the linear ADTs: arrays, stacks, and queues. Weeks 6 and 7 turn to linked representations, asking when a pointer chain beats a block of memory. Weeks 8 and 9 cover searching and sorting, with the cost of each proved rather than asserted. Weeks 10 through 12 close with the non-linear structures: hash tables, trees, and graphs. One artifact runs through the whole term: a small **expression processor** that reads $(a+b)*c$, checks its brackets, converts it, evaluates it, and looks up its identifiers. Every new structure arrives as the fix for something that program cannot yet do.

1.2 The Memory of a Running Program

Read alongside: Wirth, *Algorithms and Data Structures*, §4.1–4.2 (pp. 109, 111) — the recursive-variability argument and the pointer/anchor; Horowitz & Sahni, Ch. 1 (chapter-level).

This chapter asks one question first: *where does a running C program keep its data?* The honest answer is that a running program is given a region of memory with a simple two-part geography that this course never abandons. Figure 1.1 draws it the way every later diagram in the course will: the **stack** above for short-lived local variables, the **heap** below for memory requested at run time.

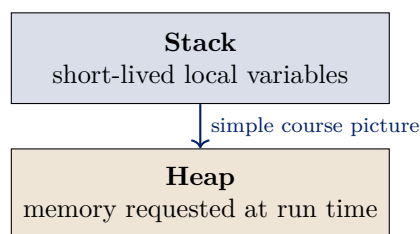


Figure 1.1: The two places a running C program keeps data: the stack for short-lived local variables, the heap for memory requested at run time. The whole course reuses this orientation.

1.2.1 The stack: automatic storage

A variable declared inside a function is created when the function starts, is available while that function is running, and disappears when the function returns. The program manages this basic lifetime automatically — no explicit allocation, no explicit cleanup. Writing `int score = 80;` inside a function is a local variable with exactly this lifetime.

To see what “automatic” really means, trace what happens to the stack as one function calls another. The deck does this with a small pair of functions: `main` holds `int a = 5;`, and calls `add(x, y)` that computes `int s = x + y;` and returns it.

Example 1.2.1 (Stack growth, Step 1: `main` starts). *When the program starts, the system creates one **frame** on the stack for `main`, holding that function’s local variables — here just `int a = 5;`. At this instant only this one frame exists (Figure 1.2). The space below it is free, waiting for the next function’s frame.*

Example 1.2.2 (Stack growth, Step 2: `add` is called). *The call `add(x=5, y=3)` pushes a second frame just below `main`’s, holding the parameters (x, y) and the local `int s = x + y;` — which evaluates to 8 (Figure 1.3). Every new function call pushes a new frame just below the current one, so each new frame sits at a **lower** address than the frame above it.*

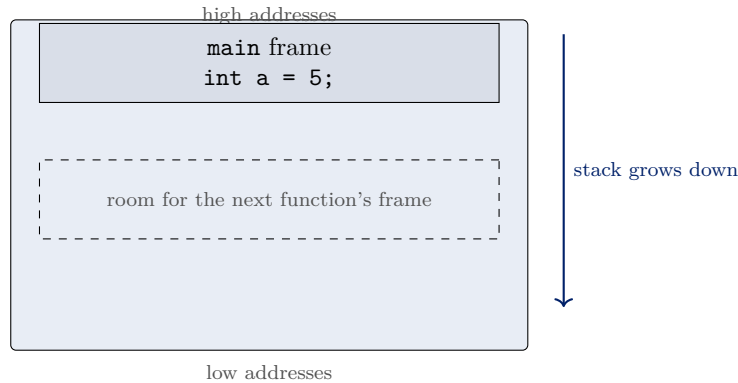


Figure 1.2: Stack growth, Step 1: `main` starts and its frame is created, holding `int a = 5;`. The stack grows *downward* — toward lower addresses.

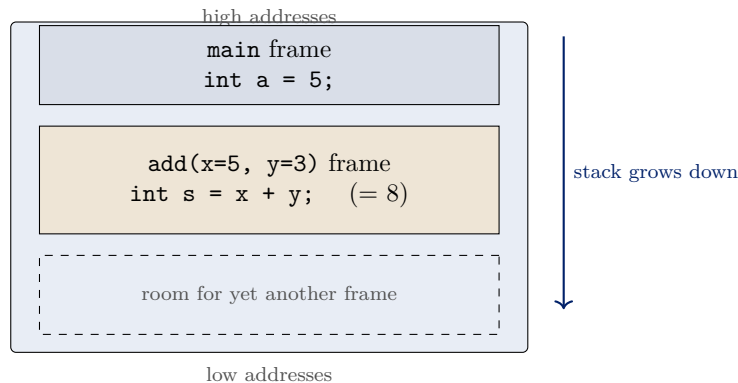


Figure 1.3: Stack growth, Step 2: the call to `add` pushes a second frame (parameters `x,y` and local `s`) at a lower address than `main`'s frame.

Example 1.2.3 (Stack growth, Step 3: `add` returns). When `add` returns, its frame is popped off the stack — it no longer exists, and its memory is available again for the next call (Figure 1.4). The answer 8 is copied back into `main`'s frame as `int s = 8;`. The stack has returned to its Step 1 shape.

The three steps together are the whole story of automatic storage: lifetime is decided by the compiler, not by you. A local variable lives exactly as long as its enclosing call. This is precisely why returning a pointer to a local variable is a bug — the frame it pointed into ceases to exist the instant the function returns.

1.2.2 Why the stack is not enough

Consider a requirement the stack simply cannot meet: the user enters the number of records only after the program starts, and we may need space for 3 records today and 300 tomorrow. The stack decides every frame's size at compile time and discards every local when its call returns. So the stack fails on both counts: we do not know the size in advance, and we may want the records to remain after the helper function that created them returns.

Guessing a maximum does not rescue us. Declaring `int a[1000000];` wastes memory whenever the true count is small, and still fails whenever it turns out to be larger. What we need is storage

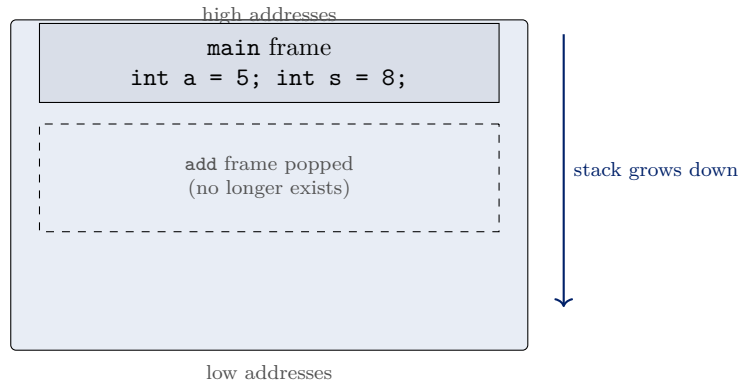


Figure 1.4: Stack growth, Step 3: `add` returns, its frame is popped and its memory is free again; 8 was copied back into `main`’s frame.

whose size is decided at run time and whose lifetime *we* control rather than the compiler. That is exactly what the **heap** provides, and the only way to reach the heap is through a pointer. Wirth puts the underlying argument crisply: recursive (variable-sized) structures “cannot be assigned a fixed amount of storage,” so the compiler cannot associate specific addresses with them; the resolution is *dynamic allocation* — allocating store to components when they come into existence during program execution — with fixed storage reserved only for the *address* of the component [3] (p. 111).

Example 1.2.4 (A prediction worth making before the vocabulary). *Two declarations:*

- `int score = 80;`
- `int *p = malloc(sizeof(int));`

Which variable can outlive the function that created it? The local variable `score` belongs to its function call and dies with it. The heap block — the one requested by `malloc` — remains until the program releases it with `free`. The pointer `p` is only the handle; the block is the data storage. Answer: the block, not the variable that names it. The exact call that makes this happen (`malloc` and `sizeof`) comes in Section 1.4.

1.3 Pointers: The Handle on Memory

Read alongside: Wirth, §4.2, p.111 — the dynamic-allocation argument and the address-as-handle picture.

Definition (Pointer). *A variable that stores an address. Its type records what kind of object resides at that address.*

An ordinary variable stores a value such as 80. A pointer stores a *place* such as “where the score lives.” The type tells C what kind of value is at that place. The declaration `int *p;` means: `p` can point to an `int` [3].

Figure 1.5 draws the distinction. The variable `p` on the left does not contain the score itself — it contains an address. The box on the right holds the actual value 80. The arrow labelled “points to” is the whole idea: `p` tells us where the score can be found.

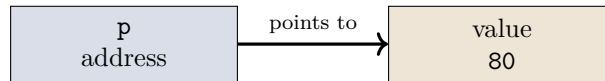


Figure 1.5: A pointer picture: `p` holds an address; the value 80 lives at that address. The pointer is the handle, not the data.

1.3.1 Pointers reach structures

Pointers earn their keep when they point at *structures*. The basic building block for most of this course is a two-field node:

```

struct node {
    int data;
    struct node *next;
};
  
```

The `data` field stores the value; the `next` field stores the address of *another* node. If there is no next node, set `next = NULL`. This is the seed of a linked list, studied properly in Week 6; here it appears only so that we can point at something real. For the field-access we prefer the arrow form: writing `p->data` reads the `data` field of the node `p` points at, and `p->next` reads its successor link. (The equivalent `(*p).data` is correct C but noisier; this course uses `->` throughout.)

1.3.2 The safe empty pointer

Definition (NULL). *The address value meaning “points at nothing.” A pointer set to `NULL` is currently empty; reading through it is undefined behaviour — in practice, a crash.*

`NULL` is the agreed way to show that a pointer is not pointing to a valid object. Use it to mark empty state, check a pointer before using the object it should point to, and never read through an empty pointer. The canonical example of its meaning: `struct node *head = NULL`; says “the list is empty.” One notation convention holds for the whole term: the null pointer is always written `NULL` in uppercase, and the list handle is always named `head`.

1.4 One Heap Block

Read alongside: Wirth, §4.2, p.111 (dynamic allocation); Horowitz & Sahni, Ch. 1 (chapter-level).

This section does one thing repeatedly: allocate exactly one block on the heap, use it, and return it. The whole memory-discipline of the course is these three moves.

1.4.1 Requesting memory: `malloc` and `sizeof`

The call `malloc(n)` asks the heap for `n` bytes of memory and returns their address, or `NULL` if the request cannot be met. You should never count those bytes by hand; the `sizeof` operator computes them for you and stays correct when the type or the target machine changes.

Example 1.4.1 (The basic pattern: one integer on the heap). `int *p = malloc(sizeof(int));`
`if (p == NULL) return 1;`
`*p = 80;`

Reading it line by line:

1. `malloc(sizeof(int))` requests from the heap exactly as many bytes as one `int` needs, and returns their address.
2. `p` stores that address. If the request failed, `malloc` returns `NULL`; the `if` guard converts that failure into a controlled exit instead of a crash later.
3. `*p = 80;` writes the value 80 into the heap block — the block at the address `p` holds. (Note the difference from the pointer picture in Figure 1.5: there `p` pointed at a pre-existing box; here the box did not exist until this call.)

The name `p` is not the heap block; the block is at the address `p` contains.

Example 1.4.2 (Allocate a complete node: the safer engineering pattern). `struct node *p = malloc(sizeof(struct node)); if (p == NULL) return 1; p->data = 42; p->next = NULL;`

This is the pattern the course will reuse every week:

1. `sizeof(struct node)` asks for exactly one node's bytes — not a hand-counted 8 or 12, which would be wrong the moment a field is added or the code is recompiled for another machine.
2. The result is checked against `NULL` before anything touches the block.
3. `p->data = 42;` stores a value in the node's `data` field.
4. `p->next = NULL;` makes the node's end explicit. The node has a clear stop: the next link is empty.

Initialising every field makes the node's state predictable; an uninitialised field would contain garbage.

1.4.2 Audit the node before you ship it

A small program is not finished when it compiles. The deck supplies a four-point review checklist that doubles as this course's allocation invariant, and it is worth internalising as a habit:

1. Was the allocation checked against `NULL`?
2. Was every field initialised before the node was used?
3. Is there exactly one clear owner responsible for `free(p)`?
4. Can the program reach the node later, or has its pointer been lost?

The engineering habit behind the list: test both the normal path and the allocation-failure path. In C, out-of-memory is not an exception the runtime raises for you; it is a `NULL` return you must notice.

1.4.3 Trace it before running it

Reading C and running C are different skills. Before running the two lines

```
int *p = malloc(sizeof(int));
*p = 80;
```

the deck asks: *after these lines, what is true?* Answer, state by state:

1. `p` contains an address — the address `malloc` returned.
2. The heap block at that address contains 80 — written there by `*p = 80;`.
3. The name `p` itself is *not* the heap block. They are two different objects in two different regions (Figure 1.6).

The challenge in the same frame: *which line would fail if `p == NULL`?* Line two. Dereferencing `NULL` with `*p` is undefined behaviour — in practice a crash. Line one would have already told you the allocation failed, if you had checked.

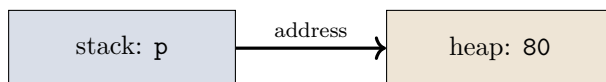


Figure 1.6: What happened after the two lines: `p` lives on the stack and holds the address of a heap block containing 80. The pointer and the block are separate objects in separate regions.

1.4.4 How the heap grows

Just as the stack picture was built from three steps, the heap gets the same treatment. The growing arrow runs *upward*: the heap grows toward higher addresses, opposite to the stack.

Example 1.4.3 (Heap growth, Step 1: one `malloc` for three integers). `int *p = malloc(3 * sizeof(int));`
`p[0] = 10; p[1] = 20; p[2] = 30;`

One `malloc` request with room for three integers grew the used heap upward, placing a block that holds 10, 20, 30 (Figure 1.7). The block survives after the allocating function returns — that is the heap’s whole point.

Example 1.4.4 (Heap growth, Step 2: a second block). `int *q = malloc(2 * sizeof(int));`
`q[0] = 40; q[1] = 50;`

*The second request places `q`’s block at a **higher** address than `p`’s (Figure 1.8). Blocks are independent: each keeps its own data, and one block’s values cannot reach into the other’s.*

Example 1.4.5 (Heap growth, Step 3: `free` returns a block). `free(p);`
`p = NULL;`

`free(p)` gives `p`’s block back to the system — it is no longer usable — and the used heap can shrink again (Figure 1.9). `q`’s block is untouched. Setting `p` to `NULL` afterwards makes the old pointer visibly empty, so it can never be mistaken for a live handle.

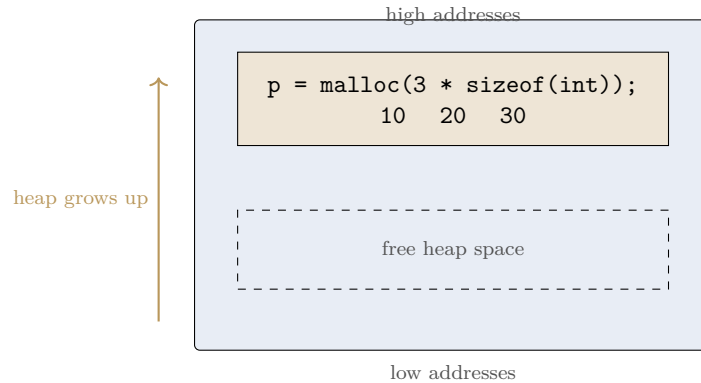


Figure 1.7: Heap growth, Step 1: `malloc(3 * sizeof(int))` places one block holding 10, 20, 30; the used heap grew upward.

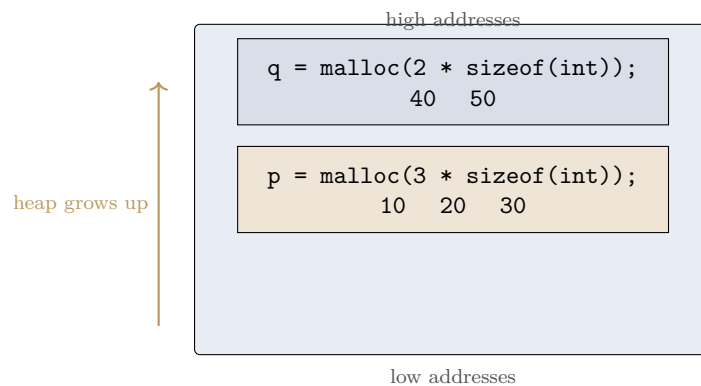


Figure 1.8: Heap growth, Step 2: a second `malloc` places `q`'s block at a higher address; the two blocks are independent.

1.4.5 Returning the memory

`free` is the counterpart of `malloc`: a block requested with `malloc` should eventually be freed. The recommended sequence is two lines that travel together:

```
free(p);
p = NULL;
```

The first line returns the heap block to the system. The second makes the old pointer visibly empty, so the handle cannot accidentally be used as though the block still existed. **Request memory, use it, return it.** This three-word rule is the entire memory-discipline of the course, and the two failures that violate it are the two beginner mistakes.

1.4.6 Two beginner mistakes

- **Memory leak:** forget to call `free`; the block stays reserved for the whole program, and if the pointer is overwritten without freeing, nothing can ever reach the block again. Nothing crashes — the program simply grows until it dies.
- **Dangling pointer:** use a pointer after its block was freed. The block is gone (or reused), so reads return garbage and writes corrupt whatever now occupies that space.

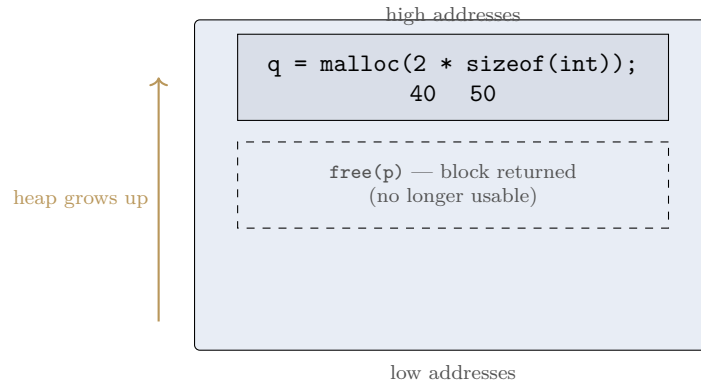


Figure 1.9: Heap growth, Step 3: `free(p)` returns `p`’s block to the system; `q`’s block is untouched.

The simple habit that prevents both: **keep track of who requested the memory and who will return it**. Decide the single owner at the moment of allocation, and that owner is responsible for exactly one `free`.

1.5 Simple Contracts

Read alongside: Sedgewick & Wayne, *Algorithms*, §1.2, p.64 — the data-type/ADT definition; §1.3, p.120 — bag as a collection type; Aho, Hopcroft & Ullman, Ch. 1 (chapter-level).

We can now build a structure and reach it through a pointer. The remaining difficulty is describing one without drowning in pointers — which brings us to the course’s central abstraction.

Definition (Abstract data type (ADT)). *A description of what we can do with data, without saying how it is stored: a promise about operations.*

Sedgewick & Wayne put it as cleanly as anyone: a data type is “a set of values and a set of operations on those values” [2] (p. 64), and the *abstract* data type is that contract stated before any implementation decides how it is met. An ADT is a promise about operations; the implementation is the code and memory arrangement behind the promise; and different implementations can follow the same promise.

Example 1.5.1 (A very small bag). *The deck’s running example is a **bag of integers** supporting exactly three operations:*

1. *`insert(x)` — put x into the bag.*
2. *`member(x)` — answer whether x is present.*
3. *`size()` — report how many items are present.*

The contract does not yet say whether the bag uses an array or a list. That is the point: the contract deliberately withholds the implementation.

1.5.1 One contract, two possible implementations

Both a dynamic array and a chain of linked nodes can honour the bag contract. Figure 1.10 draws the relationship: the contract box on top, two implementations below, arrows labelled “implements.”

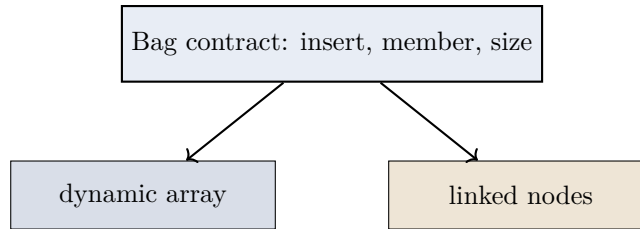


Figure 1.10: One contract, two implementations. Same operations, different internal arrangement. The caller’s code does not change when the bottom box does — that substitutability is the value of the abstraction.

The key distinction, stated once and reused all term: **same operations, different internal arrangement**. This framing runs the whole course. Weeks 3 and 7 both implement a **stack**, once over an array and once over a chain of nodes. Weeks 5 and 7 do the same for a **queue**. Week 6 and Week 10 both implement a **lookup table** for the identifiers in our expression processor, first as a list and then as a hash table. Every time, the code that *uses* the structure is written against the contract and does not care which implementation it received.

Example 1.5.2 (Classifying contract statements vs. implementation statements). *Is each statement part of the contract, or part of one implementation?*

1. “*member(7) answers yes or no.*” — *contract*. It describes what a user can ask.
2. “*The values are stored in an array.*” — *implementation*. It describes how one particular bag is built.
3. “*size() reports the number of values.*” — *contract*. It describes what a user can ask.

*The first and third describe what users can ask; the second describes how. This two-way split — what vs. how — is the habit to build now: **state the contract before writing a line of the implementation**.*

1.6 What We Cannot Yet Answer

We can now build two implementations of the same ADT, and we have no vocabulary whatever for saying which is better — only “it felt faster,” which is not an argument.

Week 2 supplies that vocabulary. **Algorithm analysis** counts an algorithm’s basic operations as a function of the input size n , then compresses that count into an asymptotic class — O , Ω , Θ — so that two implementations can be compared without ever running either of them. The bag’s two implementations differ *only* in cost; Week 2 is how we say that precisely.

Exercises

1. If **head** lives on the stack and the node lives on the heap, what exactly is lost when **main** returns without calling **free**? Who could still reach that node?
2. Draw the complete picture for these four lines, then state what is true after each one:

```

struct node *p = malloc(sizeof(struct node));
if (p == NULL) return 1;
p->data = 7;
p->next = NULL;

```

Your drawing should show which region each object lives in and the value of every field.

3. In the heap-growth trace, the second block (`q`) sits at a *higher* address than the first (`p`). If you now call `free(q)` and then `malloc` a block of three integers, what can you say about where the new block lands? (Hint: think about what “free space” the heap reuses, not about exact addresses.)
4. Classify each statement as contract or implementation, and justify: “`size()` is $\mathcal{O}(1)$ for the array bag and $\mathcal{O}(n)$ for the linked bag.” (This is a preview of Week 2 vocabulary — you are not expected to compute the bounds yet, only to place the statement on the contract / implementation split.)

Solutions

1. **What is lost:** the *reach*. `main`’s frame (which held `head`, the only pointer to the node) is popped when `main` returns, so the node’s address disappears. The node itself is still allocated on the heap — it has not been freed — but no live pointer names it any more. Nobody can reach that node; it is a memory leak. The block stays reserved for the rest of the program’s life.
2. **The complete picture:**
 - After line 1: `p` (on the stack) holds an address returned by `malloc`; a one-node block now exists on the heap, with garbage in both fields because nothing has written to it yet.
 - After line 2 (assuming the check passed): the address is known to be valid; execution continues.
 - After line 3: the heap node’s `data` field holds 7.
 - After line 4: the heap node’s `next` field holds `NULL`, marking the end of the chain. The stack holds `p`; the heap holds a node with `data = 7`, `next = NULL`.
3. **The new block’s location:** freeing `q` returns its two-integer block to the heap’s free space. A subsequent `malloc` of three integers will be served from *available* free space; the exact address is the allocator’s choice and not something the program may rely on. What you *can* say is general: the allocator reuses freed space, so the heap’s used region can shrink and later regrow — the important invariant is that each block is reached only through a live pointer, never through a guess about its neighbour.
4. **Contract or implementation:** the statement is *implementation* — it names specific data structures (the array bag, the linked bag) and specific costs. The contract would say only “`size()` reports the number of items,” saying nothing about how fast. Cost claims belong to implementations; the contract deliberately stays silent so that the two can be compared at all.

Summary

- A program is **algorithms** plus **data structures**; structured programming disciplines only the first.

- **Stack** storage is automatic and scope-bound: a frame is pushed on call and popped on return, and the stack grows *down*. **Heap** storage is run-time sized and lifetime-controlled by you: `malloc` obtains it, `free` returns it, and the heap grows *up*.
- A **pointer** stores an address — a handle on a value or structure, never the object itself. `NULL` means “points to nothing,” and dereferencing it is undefined behaviour.
- The allocation idiom: `malloc` with `sizeof`, check the result against `NULL`, initialise every field, use the block, then `free` it and set the pointer to `NULL`.
- The two silent failures: **memory leak** (never freed) and **dangling pointer** (used after being freed). Every allocation gets exactly one owner.
- An **ADT** is a contract about operations; a **data structure** is one implementation of it. State the contract before writing the code, and the rest of the course compares implementations of shared contracts.

References

- [1] Ellis Horowitz and Sartaj Sahni. *Fundamentals of Data Structures*. Galgotia Booksources, 2nd edition, 2008.
- [2] Robert Sedgewick and Kevin Wayne. *Algorithms*. Addison-Wesley, 4th edition, 2011.
- [3] Niklaus Wirth. *Algorithms and Data Structures*. Author (freely distributed), 2004. Oberon version, August 2004; revision of the 1985 Prentice-Hall edition. Page numbers cited in CS301 material refer to this Oberon PDF, not to the 1985 print edition.